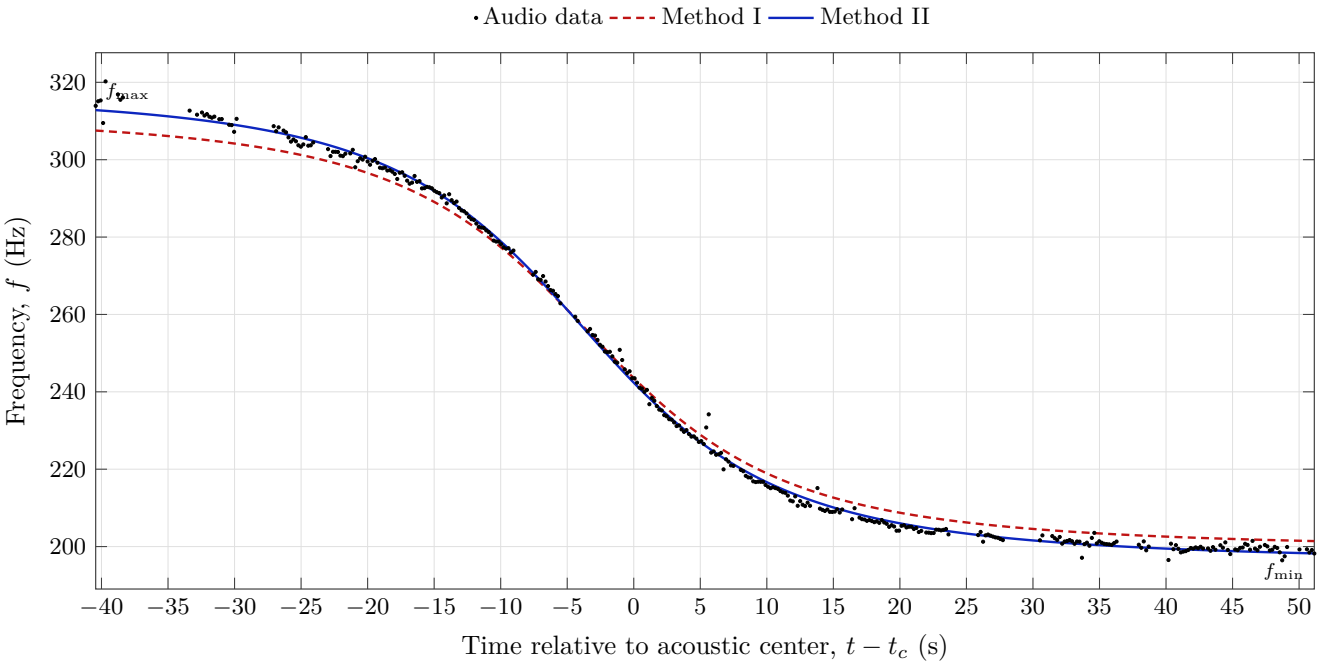
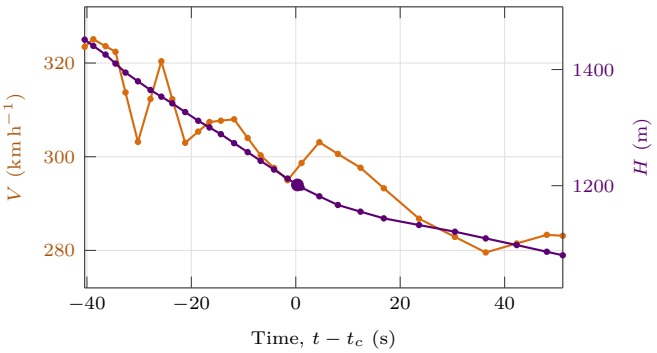
R19: N4004R

UTC audio ridge and N4004R-track-EGM96.kml

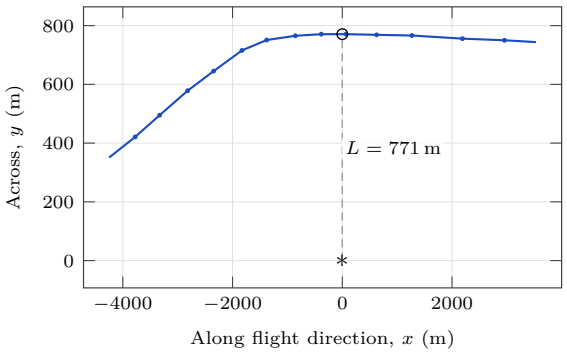
V constant H varying; analyzed; audio data with Method I and Method II fitted curves



ADS-B speed and height



Ground track



Reference / method	V (km h^{-1})	$R_{\min}$ (m)	L (m)	H (m)
ADS-B	300.31	1427.38	770.89	1201.32
Method I	270.00 ± 1.418	1205.35 ± 44.975	770.89	926.61 ± 58.505
Method II	290.61 ± 0.933	1314.52 ± 15.231	770.89	1064.75 ± 18.804

L is the ground-track offset from ADS-B; the same value is used in all three rows. Each height follows from $H = \sqrt{R_{\min}^2 - L^2}$, so a real height needs $R_{\min} > L$. A dash means $R_{\min} \le L$. For ADS-B, $R_{\min} = \sqrt{H^2 + L^2}$.

Method I guides: $f_{\max} = 311.995$ Hz (first 18 points) and $f_{\min} = 199.226$ Hz (last 18 points).